

P-hacking with Random Data Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 35-45 min

Category	Natural Sciences	Time	35-45 min
Difficulty	Extended	ID	p-hacking-with-random-data

Big idea: Methods can create false confidence.

Student-Facing Prompt

Inspect P-hacking with Random Data Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

P-hacking with Random Data Stimulus A	Student-facing stimulus 1: A random dataset where students hunt for any impressive correlation. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
P-hacking with Random Data Stimulus B	Student-facing stimulus 2: Multiple subgroup comparisons that eventually produce a 'significant' pattern. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
P-hacking with Random Data Reveal / Twist	Student-facing stimulus 3: A publication pitch followed by a methods audit. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Methods can create false confidence.

Actual Lesson Questions

#	Question for students
1	After inspecting P-hacking with Random Data Stimulus A, what is your first knowledge claim?
2	Which exact detail from P-hacking with Random Data Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Can correct methods lead to false knowledge?

Ready-to-Use Examples

- A random dataset where students hunt for any impressive correlation.
- Multiple subgroup comparisons that eventually produce a 'significant' pattern.
- A publication pitch followed by a methods audit.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Give groups a random data table.
2. Let them test as many relationships as possible.
3. Reward groups for finding an apparently interesting pattern.
4. Reveal that, with enough searches, chance findings become likely.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can correct methods lead to false knowledge?
- Why are preregistration, replication and transparency important?
- How can a result be statistically impressive but epistemically weak?

Knowledge Questions

- Can correct methods lead to false knowledge?
- Why are preregistration, replication and transparency important?
- How can a result be statistically impressive but epistemically weak?

Transfer Example

For example, run this activity with a random dataset where students hunt for any impressive correlation. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...